

# THE INTUITION OF THE CENTRAL LIMIT THEOREM VIA PYTHON

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## EVERYTHING IS NORMAL

In Article No.14, we explored the Law of Large Numbers, which shows that sample averages converge to their expected value as the sample size increases. While this result explains where the sample mean converges to, it tells us little about the distribution of the sample mean itself. The Central Limit Theorem answers this question and it wouldn't be an overstatement to say that it serves as a bridge between the gap of probability and statistics.

The Central Limit Theorem states that the distribution of a suitably standardised sample mean approaches a normal distribution as the sample size becomes large, regardless of the underlying distribution. It forms the theoretical foundation of many statistical techniques, including confidence intervals, hypothesis testing, and Monte-Carlo simulations.

The origins of the Central Limit Theorem can be traced back to a famous mathematician, Abraham de Moivre, in 1733. He discovered that the Binomial distribution, arising from repeated fair coin tosses, could be approximated by a Normal distribution when the number of trials became large. This result was later generalised by Pierre-Simon Laplace, leading to what is often referred to as the De Moivre-Laplace Theorem. Perhaps the most remarkable consequence of this work is that the Normal distribution emerges naturally from the aggregation of many small random effects. This phenomenon helps explain why quantities such as human heights, measurement errors, examination scores, and many financial variables often exhibit approximately normal behaviour.

## THE CENTRAL LIMIT THEOREM

Let  $X_1, X_2, \dots, X_n$  be a sequence of IID random variables with finite mean,  $\mathbb{E}[X_i] = \mu$  and finite variance,  $\text{Var}(X_i) = \sigma^2$ . The **Central Limit Theorem** states that as the sample size,  $n$  becomes large, the distribution of the standardised sample mean  $\tilde{S}_n = \frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$  converges in distribution to a standard normal random variable. That is,  $\tilde{S}_n \xrightarrow{d} N(0, 1)$ .

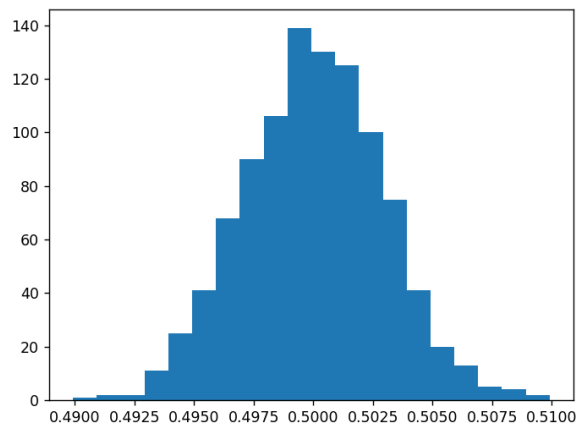
In words, the implication is that regardless of the original distribution of the data, the distribution of sample means becomes increasingly Normal as the sample size grows. In other words, if we repeatedly draw samples from almost any distribution and compute their means, the resulting histogram of sample means will eventually resemble the bell-shaped curve. It is important to note that the Central Limit Theorem concerns the distribution of the sample mean, not the distribution of the original random variable itself. For example, a Uniform, Exponential, or even a highly skewed distribution need not be Normal. Nevertheless, the distribution of their sample means will become approximately Normal if the sample size is sufficiently large.

## UTILISING PYTHON

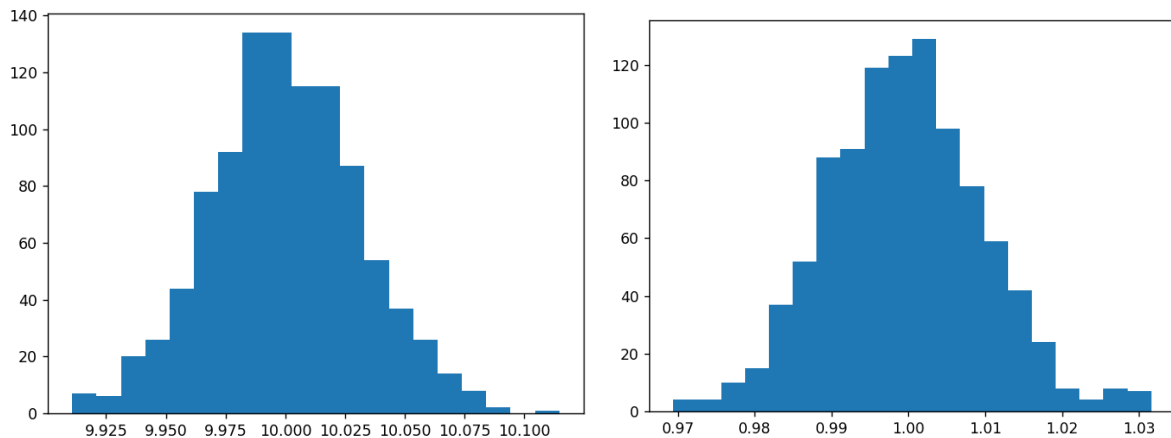
Like the Law of Large Numbers, the intuition behind the Central Limit Theorem is best developed through simulation. The following Python code provides a simple illustration of the theorem in action:

```
1 # Import modules
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import scipy
5
6 Stilden = [] # Create empty array to store sample means
7 for i in range(1000):
8     S = np.random.uniform(0, 1, 10000) # Generate random numbers from a uniform distribution
9     xbarn = np.mean(S) # Calculate mean of the sample
10    Stilden.append(xbarn) # Store sample mean in the array
11 plt.hist(Stilden, bins=20) # Plot histogram of sample means
12 plt.show() # Display histogram
```

The resulting output of the uniform distribution is:



We can also investigate other underlying distributions by making a small modification to the code. Replacing the sampling statement with " $S = np.random.poisson(10, 10000)$ " or " $S = np.random.exponential(1, 10000)$ ", allowing us to generate samples from a Poisson and Exponential distribution, respectively. The resulting outputs are shown below.



Despite the underlying distributions being very different, the distribution of the sample means remains approximately Normal, providing a clear demonstration of the Central Limit Theorem. We could also investigate how quickly the Central Limit Theorem is implemented. By varying the sample size  $n$ , we can observe how the distribution of the sample mean evolves and gradually approaches a Normal distribution.

## CONCLUSION

The Central Limit Theorem is one of the most impactful theorems in probability theory, explaining why the Normal distribution appears so frequently in practice. Through simulation, we have seen how sample means converge towards a bell-shaped distribution regardless of the underlying data.

ALEXANDER FLANN